Lecture 1: The mean, the median, and the mode.

(Link: `https://youtu.be/-14BImgVENA` **)**

This video is a 13-second musical jingle. The entire content is:

> "The mean, the median, and the mode. When it's normal, they're the same."

Lecture 2: The Exponential Distribution

(Link: `https://youtu.be/wtLQUjfomig` **)**

This video is a 16-second musical jingle. The entire content is:

> "The exponential distribution is useful when you're estimating time between two events, like the amount of time that passes between two StatQuest videos."

Lecture 3: Population and Estimated Parameters, Clearly Explained!!!

(Link: `https://youtu.be/vikkiwjQqfU` **)**

Here are the complete, detailed notes for the 14-minute lecture.

Lecture Notes: Population and Estimated Parameters

I. The Core Idea: Population vs. Sample

The main goal of statistics is to understand a **population**, but we can usually only collect data from a **sample**.

- **Population:** This represents *every single item* of interest.
  - *Example:* All 240 billion liver cells in a human liver.
  - *Example:* Every single store in a specific grocery chain.
- **Sample:** A small subset of the population that we *actually measure*.
  - *Example:* Measuring mRNA in just 5 liver cells.

We rarely, if ever, have enough time or money to measure the entire population. So, we must use the sample to learn about the population.

II. Population Parameters (The "Truth")

If we *could* measure the entire population, we could create a perfect histogram of the data. This histogram can be approximated by a smooth curve, called a **statistical distribution** (e.g., a Normal Distribution).

The values that define the exact shape and location of this "true" distribution are called **Population Parameters**.

- For a **Normal Distribution**, the two parameters are:
  1. **Population Mean (μ):** The true average, located at the center of the curve.
  2. **Population Standard Deviation (σ):** The true spread, which defines the width of the curve.

These "true" parameters are what we want to know because they allow us to calculate true probabilities (e.g., the exact probability of a cell having more than 30 transcripts). Other distributions have different parameters (e.g., an Exponential distribution is defined by a `rate` parameter).

III. Estimated Parameters (The "Guess")

Since we can't get the *true* population parameters, we use our *sample* to calculate **Estimated Parameters**.

- We use the mean of our 5-cell sample to *estimate* the true population mean (μ).
- We use the standard deviation of our sample to *estimate* the true population standard deviation (σ).

IV. The Problem of Reproducibility

This creates a fundamental problem:

- In our first experiment (a sample of 5 cells), our estimated mean might be **17.6**.
- If we repeat the experiment (a *new* sample of 5 cells), our estimated mean might be **19.2**.
- Both of these are different from the *true* population mean (e.g., 20).

If every experiment gives a different estimate, how can our results be **reproducible**?

V. The Solution: Sample Size and Confidence

This is the most important part of statistics. The solution comes from two concepts:

1. Sample Size (More Data is Better)

- The **more data** we have (a larger sample size), the **better** our estimates become.
- *Example:*
  - With 2 measurements, our estimated mean was 11 (very far from the true mean of 20).
  - With 5 measurements, our estimated mean was 17.6 (closer to 20).
  - With 10 measurements, our estimate would be even better.

2. Quantifying Confidence

- Statistics allows us to quantify *how much confidence* we have in our estimates.
- We use powerful tools like **p-values** and **confidence intervals** for this.
- These tools tell us if the difference between two experiments (like 17.6 vs. 19.2) is "statistically significant" or just due to random chance.
- If the difference is *not* significant, we can conclude that our results are **reproducible**, even though the exact numbers are slightly different.

Summary

1. A **population** is every single item; a **sample** is a small subset.
2. **Population parameters** (like μ and σ) are the "true" values we want to know.
3. We use our sample to *estimate* these parameters.
4. We use p-values and confidence intervals to **quantify our confidence** in those estimates.
5. **More data (larger sample) = More confidence**.
6. This process (estimating parameters and quantifying confidence) is what allows science to be reproducible.